

MA3265 Number Theory Cheatsheet

1 Primes and Basic Modular Arithmetic

1.1 Complete and Reduced Residue Systems

- If $\{r_1, r_2, \dots, r_n\}$ is a CRS modulo n and $\gcd(a, n) = 1$, then $\{ar_1 + b, ar_2 + b, \dots, ar_n + b\}$ is also a CRS modulo n for any integer b
- If $\{r_1, r_2, \dots, r_{\phi(n)}\}$ is a RRS modulo n and $\gcd(a, n) = 1$, then $\{ar_1, ar_2, \dots, ar_{\phi(n)}\}$ is also a RRS modulo n

1.2 Wilson's Theorem

$(p-1)! \equiv -1 \pmod{p}$ iff p is prime.

1.3 Euler's Theorem

- If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$, where $\phi(n)$ is Euler's totient function.
- If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$

2 Solving System of Congruences

2.1 Chinese Remainder Theorem (CRT)

For general systems:

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2} \\&\vdots \\x &\equiv a_k \pmod{m_k}\end{aligned}$$

where $\gcd(m_i, m_j) = 1$ for $i \neq j$:

1. Let $M = m_1 m_2 \cdots m_k$
2. Let $M_i = M/m_i$
3. Find M_i^{-1} such that $M_i \cdot M_i^{-1} \equiv 1 \pmod{m_i}$
4. The solution is $x \equiv \sum_{i=1}^k a_i M_i M_i^{-1} \pmod{M}$

2.2 Solving Linear Congruences

For $ax \equiv b \pmod{m}$:

- Has a solution if and only if $\gcd(a, m) \mid b$
- If $\gcd(a, m) = 1$, there is a unique solution modulo m
- If $\gcd(a, m) = d > 1$ and $d \mid b$, there are d distinct solutions modulo m

2.3 Primitive Roots

- An integer g is a **primitive root** modulo n if $\text{ord}_n(g) = \phi(n)$
- **Theorem:** Primitive roots exist modulo n iff $n = 1, 2, 4, p^k, 2p^k$ for odd prime p

2.4 Solving $x^m \equiv c \pmod{n}$

1. Break into system of equations modulo various prime powers
2. Solve the ones with odd prime powers p^k with primitive roots
3. Every integer modulo 2^k can be written uniquely as $\pm 5^j$ for $0 \leq j < 2^{k-2}$
4. Glue solutions with CRT.

2.5 Hensel's Lifting Lemma

Given a_k with $f(a_k) \equiv 0 \pmod{p^k}$, compute:

$$a_{k+1} = a_k - \frac{f(a_k)}{p^k} \cdot (f'(a_k))^{-1} \cdot p^k \pmod{p^{k+1}}$$

3 Quadratic Residues

3.1 Properties of Quadratic Residues

For an odd prime p and integer a with $\gcd(a, p) = 1$:

- $a \in \text{QR}(p^e)$ if and only if $a \in \text{QR}(p)$ for any $e \geq 1$
- $\text{QR}(p) = \{1^2, 2^2, \dots, (\frac{p-1}{2})^2\}$
- $\text{QR}(p) = \{u^{2k} : k \in \mathbb{Z}\}$ where u is a primitive root

3.2 Legendre Symbol

For an odd prime p and integer a :

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a QR modulo } p \\ -1 & \text{if } a \text{ is a NQR modulo } p \\ 0 & \text{if } p \mid a \end{cases}$$

3.3 Properties of Legendre Symbol

- $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$ (Euler's Criterion)
- $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right)$ (Multiplicative)

3.4 Quadratic Reciprocity

For distinct odd primes p and q :

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

i.e., they are equal when both $p, q \equiv 3 \pmod{4}$.

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} 1 & \text{if } p \equiv 1, 7 \pmod{8} \\ -1 & \text{if } p \equiv 3, 5 \pmod{8} \end{cases}$$

4 Pythagorean Triples

4.1 Definition

A Pythagorean triple is a set of three positive integers (a, b, c) satisfying $a^2 + b^2 = c^2$. A primitive Pythagorean triple has $\gcd(a, b, c) = 1$.

4.2 Parametrization of Primitive Pythagorean Triples

All primitive Pythagorean triples (a, b, c) can be generated by:

$$\begin{aligned}a &= s^2 - t^2 \\b &= 2st \\c &= s^2 + t^2\end{aligned}$$

where $s > t > 0$, $\gcd(s, t) = 1$, and s and t have opposite parity (one even, one odd).

4.3 Chord-and-Tangent Method

We observe that finding integer solutions to $a^2 + b^2 = c^2$ is equivalent to finding rational points on $x^2 + y^2 = 1$.

- Pick an (obvious) rational point P (e.g., $(1, 0)$ for the unit circle).
- Draw a secant line with rational slope through this point
- Find the second intersection with the conic
- This gives a parametrization of all rational points

$$(x, y) \rightarrow \left(\frac{s^2 - t^2}{s^2 + t^2}, \frac{2st}{s^2 + t^2}\right)$$

4.4 Congruent Numbers

- A positive rational number n is a **congruent number** if it is the area of a right triangle with rational sides
- Equivalently, n is congruent if and only if the elliptic curve $y^2 = x^3 - n^2x$ has a rational point with $y \neq 0$

$$\begin{aligned}(a, b, c) &\rightarrow \left(\frac{nb}{c-a}, \frac{2n^2}{c-a}\right) \\ \left(\frac{x^2 - n^2}{y}, \frac{2nx}{y}, \frac{x^2 + n^2}{y}\right) &\leftarrow (x, y)\end{aligned}$$

4.5 Congruum Numbers

- n is congruum if there exists $x \in \mathbb{Z}$ such that $x^2 - n, x^2, x^2 + n$ are all square numbers

$$\begin{aligned}(a, b, c) &\rightarrow \left(\frac{b-a}{2}, \frac{c}{2}, \frac{b+a}{2}\right) \\ (t-r, t+r, 2s) &\leftarrow (r, s, t)\end{aligned}$$

5 Pell's Equations and Continued Fractions

5.1 Pell's Equation

- Standard form: $x^2 - dy^2 = 1$ where d is a positive non-square integer
- Generalized form: $x^2 - dy^2 = n$ for $n \neq 0$
- If (x_1, y_1) is the smallest positive solution to $x^2 - dy^2 = 1$, then all solutions are given by (x_k, y_k) where:

$$x_k + y_k\sqrt{d} = (x_1 + y_1\sqrt{d})^k$$

5.2 Continued Fractions

Simple continued fraction: $[a_0; a_1, a_2, \dots] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}}$

5.3 Continued Fraction of $\sqrt{d}$

- For a non-square integer $d > 0$, $\sqrt{d} = [a_0; \overline{a_1, a_2, \dots, a_n}]$ (periodic)
- $a_0 = \lfloor \sqrt{d} \rfloor$
- Process for finding continued fraction:

$$\begin{aligned}\alpha_0 &= \sqrt{d} \\ a_0 &= \lfloor \alpha_0 \rfloor \\ \alpha_{i+1} &= \frac{1}{\alpha_i - a_i} \\ a_{i+1} &= \lfloor \alpha_{i+1} \rfloor\end{aligned}$$

5.4 Convergents

- The n -th convergent of $[a_0; a_1, \dots]$ is $\frac{p_n}{q_n}$ where:

$$p_0 = a_0, \quad p_1 = a_0a_1 + 1, \quad p_n = a_n p_{n-1} + p_{n-2}$$

$$q_0 = 1, \quad q_1 = a_1, \quad q_n = a_n q_{n-1} + q_{n-2}$$

- Properties:

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$$

$$\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} = \frac{(-1)^{n-1}}{q_n q_{n-1}}$$

- For an irrational $\alpha = [a_0; a_1, \dots]$:

$$\frac{p_{2k}}{q_{2k}} > \alpha > \frac{p_{2k+1}}{q_{2k+1}}$$

That is, even-indexed convergents approach α from above (decreasing), while odd-indexed convergents approach from below (increasing).

5.5 Approximation by Convergents

- For any irrational α , there exist infinitely many rational numbers $\frac{p}{q}$ such that:

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{2q^2}$$

Then, $\frac{p}{q}$ must be a continued fraction convergent of α .

- Hurwitz's Theorem: For any irrational number α , there

exist infinitely many rational numbers $\frac{p}{q}$ such that:

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5}q^2}$$

This bound is sharp, as demonstrated by the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$.

5.6 Solving Pell's Equation Using Continued Fractions

- For $x^2 - dy^2 = 1$:
 - If the period length of $\sqrt{d}$ is even, the fundamental solution is (p_{n-1}, q_{n-1}) where n is the period length
 - If the period length is odd, the fundamental solution is (p_{2n-1}, q_{2n-1})
- For $x^2 - dy^2 = -1$:
 - Has a solution if and only if the period length of $\sqrt{d}$ is odd
 - The fundamental solution is (p_{n-1}, q_{n-1}) where n is the period length

5.7 Examples

- $\sqrt{2} = [1; \overline{2}]$
 - Convergents: $\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \dots$
 - Fundamental solution to $x^2 - 2y^2 = 1$: $(3, 2)$
- $\sqrt{7} = [2; \overline{1, 1, 4}]$
 - Fundamental solution to $x^2 - 7y^2 = 1$: $(8, 3)$

5.8 Solving Generalized Pell's Equations

Theorem: For $x^2 - dy^2 = n$ where $n \neq 0$:

1. Find the fundamental solution (a, b) to $x^2 - dy^2 = 1$ where $a, b > 0$
2. Define $u = a + b\sqrt{d}$
3. Every integral solution is of the form (x_k, y_k) where:

$$x_k + y_k\sqrt{d} = (x_0 + y_0\sqrt{d})u^k, \quad k \in \mathbb{Z}$$

where (x_0, y_0) is a solution satisfying the bounds:

$$|x_0| \leq \sqrt{|n|} \cdot \frac{\sqrt{u} + \frac{1}{\sqrt{u}}}{2}$$

$$|y_0| \leq \sqrt{|n|} \cdot \frac{\sqrt{u} - \frac{1}{\sqrt{u}}}{2\sqrt{d}}$$

6 Binary Quadratic Forms

6.1 Definition

A binary quadratic form (BQF) is $f(x, y) = ax^2 + bxy + cy^2$ where $a, b, c \in \mathbb{Z}$.

6.2 Discriminant

The discriminant of $f(x, y) = ax^2 + bxy + cy^2$ is $\Delta = b^2 - 4ac$.

6.3 Representation of Integers

- n is represented by f if there exist integers x, y such that $f(x, y) = n$
- n is properly represented if $\gcd(x, y) = 1$
- If n is a nonzero integer, then there exists a BQF of Δ that properly represents n iff Δ is a QR mod $4|n|$.
- If p is an odd prime, then there exists a BQF of Δ that represents p iff Δ is a quadratic residue modulo p .

6.4 Reduced Forms

A positive definite form $f(x, y) = ax^2 + bxy + cy^2$ is reduced if:

- $|b| \leq a \leq c$
- If $|b| = a$ or $a = c$, then $b \geq 0$

Key Properties: For reduced form (a, b, c) with discriminant Δ :

- $\Delta < 0 \Rightarrow a, c$ have same sign and $|a| \leq \sqrt{-\Delta/3}$
- $\Delta > 0 \Rightarrow a, c$ have opposite signs and $|a| < \sqrt{\Delta/2}$
- Every $\text{SL}_2(\mathbb{Z})$ orbit contains at least one reduced form
- The reduced form is only unique if $\Delta < 0$.
- S_Δ contains finitely many equivalence classes of forms

6.5 Reduction Algorithm

To reduce a form (a, b, c) with discriminant Δ :

- Use transformations represented by matrices:

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} : (a, b, c) \mapsto (c, -b, a)$$

$$T^m = \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} : (a, b, c) \mapsto (a, b + 2am, am^2 + bm + c)$$

- Reduction steps:

1. If $b \notin (-|a|, |a|]$: apply T^m with $m = -\lfloor \frac{b}{2a} \rfloor$
2. If $b < 0$ or $|c| < |a|$: apply S

6.6 Dirichlet Composition

- For primitive forms $f_1 = (a_1, b_1, c_1)$ and $f_2 = (a_2, b_2, c_2)$ with same discriminant Δ and $\gcd(a_1, a_2, \frac{b_1+b_2}{2}) = 1$, their composition $f_3 = (a_3, b_3, c_3)$ is defined by:

$$a_3 = a_1 a_2$$

$$b_3 \equiv b_1 \pmod{2a_1}, \quad b_3 \equiv b_2 \pmod{2a_2},$$

$$b_3^2 \equiv \Delta \pmod{4a_1 a_2}$$

$$c_3 = \frac{b_3^2 - \Delta}{4a_3}$$

- The class number $h(\Delta)$ is the number of equivalence classes of primitive forms with discriminant Δ
- $\text{Cl}(\Delta)$ forms a finite abelian group under composition
- For $\Delta < 0$, order of $\text{Cl}(\Delta)$ is $h(\Delta)$